Why do we study the geometry of surfaces?

R^3 R^3 R^3 S p p p K S_p **Theorema Egregium** K $S \subset R^3$ T_pS $p \in S$ $N(p)$ $N(p)$ $S^2 \subset R^3$ $S \subset R^3$ $N : S \to S^2$

$N : S \longrightarrow S^2, p \longmapsto N(p)$

Gauss

$N : S \to S^2$

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$dN_p : T_pS \to T_{N(p)}S^2$ p $N(p)$ $T_{N(p)}S^2$ T_pS $N(p) dN_p$ T_pS $N : S \to S^2$ p

Weingarten

$S_p = -dN_p : T_pS \longrightarrow T_pS . 3-2-1$

S_p

$\langle S_p(v), w\rangle = \langle v, S_p(w)\rangle, \forall\, v, w \in T_pS . 3-2-2$

S_p

dN_p

$dN_p = 0$ $S_p = 0$ $dN_p(v) = -v$ $S_p = I$ $dN = 0$ $dN = -I$ $S = I$ S_p I $S \subset R^3$ $S_p : T_pS \to T_pS$ p

$_p(v, w) = \langle S_p(v), w\rangle = \langle -dN_p(v), w\rangle, v, w \in T_pS . 3-2-3$

I

$S_p =$
I
$dN_p(v) =$
$-v$
$S_p =$
$-dN_p =$
I
$k_1 \equiv$
$k_2 \equiv$
1
$z =$
x^2+
y^2
$\mathbf{x}(u, v) =$
$(u, v, u^2+$
$v^2)$

$$L = \frac{2}{\sqrt{1+4(u^2+v^2)}}, M = 0, N_{coeff} = \frac{2}{\sqrt{1+4(u^2+v^2)}}.3-2-7$$

$L =$
$N_{coeff} =$
$2M =$
0
$-N$.
L, M, N_{coeff}
$_ocr/images/53c3f147d73fbbaff460736e7109c90902470b480130ac9116222c530a8f7b29.jpg$
$v, w \in$
T_pS

$$_p(v, w) = \langle S_p(v), w \rangle = I_p(S_p(v), w).3-2-8$$

E, F, G
S_pS_p
$S_p = (\ E\ ) FFG^{-1} (\ L\ ) MMN_{coeff}.3-2-93-7$

$C \subset$
S
S
$K >$
0
k
C
p
$|k| \geq \min(|k_1|, |k_2|),$

k_1
k_2
S
p
S
$|k_1| \leq$
1
$|k_2| \leq$
1
k
S
$|k| \leq$
1
H
$p \in$
S
$$H = \frac{1}{\pi}\int_0^{\pi} k_n(\theta) d\theta,$$

$k_n(\theta)$
p
θ
p
$z =$
x^2+
y^2
x^2+
y^2-
$z^2 =$
1
x^2+
$y^2 =$
$\cosh^2 z$
$N \circ$
α
$N : S \to$
S^2
$\alpha : I \to$
S
S^2
α
$C =$
$\alpha(I)$
k
p
$k = |k_n k_N|,$

k_n
p